বাংলা পাঠাগার: website, ebook reader, and book-creation plan

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Status: Planning document. Implementation and acceptance remain outstanding.

Delivery order: Website → standalone ebook reader → intelligent curation and book-creation engine.

This document brings the reported problems, additional investigation areas, product decisions, implementation work, and acceptance criteria into one place. Earlier implementation reports are historical evidence; they do not establish acceptance of this revision. Existing working-tree changes must be checked against this plan before being marked complete.

The objective is a complete, consistent product with usable admin and reader workflows, followed by a source-grounded engine that understands book structure and bibliographic credits. Passing a build or loading a page is not sufficient evidence of completion.

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1. Decisions and boundaries

Decision	Required behavior
Three independent services	Website, reader, and creation engine have independent builds, configuration, health checks, tests, and releases, in the same repository.
No spending	No paid APIs, subscriptions, purchased hosting, or additional hardware. Use available infrastructure and local inference. Do not fall back to paid services.
Implementation order	Finish the website and reader acceptance gates before implementing the intelligent engine. Define their interfaces early to prevent rework.
Languages	Bengali and English throughout the interface. Preserve book text, titles, and credited names in their original language.

Decision	Required behavior
Source	Automated creation uses one configured source website. Its identity belongs in configuration, not provider-specific names in application code or navigation.
Manual import	Keep EPUB upload. Remove printed-book creation from the interface. Preserve existing physical-book records and personal reading logs.
Publication	Automatically publish website-sourced books after all required checks pass. Do not impose routine manual editorial review.
Incomplete source books	The source website's unfinished classification is separate from collection, extraction, generation, or application defects.
Administration	Admin users primarily enter Admin and can switch to their personal Reader View with the same account. Switching view never changes permissions.
Time	Use Asia/Dhaka and a 12-hour clock, with correct localized dates, digits, and day periods.
Presentation	No underlined navigation or decorative heading counters. The logo includes বাংলা পাঠাগার without a repeated adjacent name.
Data preservation	Preserve account and book IDs, entitlements, source evidence, previous editions, notes, highlights, bookmarks, and progress.

New dependencies remain subject to AGENTS.md. The local model and runtime below are proposed dependencies; this document does not install them or authorize paid resources.

What completion means

Every audited requirement must have an implementation result and verification evidence. Release requires no open visual or UX defects in the agreed inventory and passing functional, security, and data-preservation checks. An issue must not be closed because a screenshot was replaced, an assertion weakened, an error hidden, or a difficult book skipped.

This is a measurable release standard. It is not a mathematical guarantee that an undiscovered defect can never exist.

2. Three independent services

Service	Owns	Interfaces and boundaries
Website	Public catalog, accounts, permissions, personal library, annotations, Kindle workflows, publication records, and admin interface.	Existing authorized public/private APIs; admin commands forwarded to the engine; validated edition acceptance.

Service	Owns	Interfaces and boundaries
Ebook reader	Reading interface, EPUB rendering, typography, navigation, search, fullscreen, gestures, and device-local EPUB reading.	Website APIs for authorized sessions, resources, progress, and annotations. No separate account database.
Curation and creation engine	Source discovery, collection, snapshots, entity extraction, structural understanding, jobs, schedules, validation, generation, and update detection.	Private versioned command/status/artifact interfaces and idempotent publication handoff to the website.

Integration requirements

- Keep stable public URLs. Route reader and local-reader pages to the reader service through the same-origin proxy; give reader build assets a distinct prefix.
- Use full document navigation at service boundaries so one application's client router does not attempt to render another application's pages.
- Retain website session authentication and account isolation. Do not introduce bearer credentials into reader URLs.
- Give the engine its own logical database and storage ownership. Reference website accounts and books through stable IDs and revisioned contracts, not cross-service ORM joins.
- Authenticate private service calls with scoped, short-lived credentials. Recheck current authority at protected processing boundaries and before publication.
- Keep existing public APIs compatible while introducing private engine contracts. Generate client types from the actual schemas.
- Transfer immutable artifacts with checksums. Stage a replacement edition, verify it, then atomically activate it in the website.
- Use durable outbox/inbox records and idempotency keys so retries cannot create duplicate jobs, books, or publications.
- Share existing infrastructure where appropriate. Independently deployable product services can contain API, worker, scheduler, and inference processes.
- Preserve old checkpoints and queue messages during migration. Do not purge queues, reset data, or enable external delivery as part of this redesign.

3. Issue register and audit method

Initial issue register

“Reported” means the user encountered the problem and current behavior needs rechecking. “Confirmed” refers to inspected implementation or planning-time evidence. Neither status implies that an earlier attempted fix has been accepted.

ID	Issue or concern	Evidence/status	Workstream
WEB-01	Different pages use competing control, typography, spacing, and alignment rules.	Confirmed styling structure; repeatedly reported visually.	Shared design system.
WEB-02	Header wraps or breaks; admin and general navigation are not consistently distinguished.	Reported; existing changes need fresh acceptance.	Both shells and responsive navigation.
WEB-03	Important admin actions and destinations are hard to discover.	Reported.	Navigation and task hierarchy.
WEB-04	Cards have unequal heights and inconsistent metadata/action placement.	Confirmed wrapper/stretch problem; attempted fix remains unverified.	Shared book cards.
WEB-05	Search, sort, and Filter controls have different heights and font metrics.	Confirmed rendered measurements.	Shared browse controls.
WEB-06	Export headings, buttons, and instructions occupy disproportionate space.	Confirmed component structure; compact disclosure needs acceptance.	Export controls.
WEB-07	Filters include unwanted book-type and added/request-date fields or excessive empty space.	Reported; search saved filters and old URLs too.	All browsing and administrative lists.
WEB-08	Notes has broken alignment, spacing, or card layouts.	Reported; previous repairs require current-state verification.	Notes views, toolbar, and annotation cards.
WEB-09	Automation and Users & Access lack coherent layout and clear workflows.	Reported; previous repairs require current-state verification.	Administrative lists and forms.
WEB-10	Unfinished Books misses the compact source-page container and has a confusing hierarchy.	Confirmed container mismatch; attempted fix remains unverified.	Unfinished workspace.
WEB-11	Book details do not reliably expose covers, Read, Save, Kindle, HTML, and EPUB actions.	Reported; test ready, unavailable, and denied states independently.	Book details and list actions.
WEB-12	Pagination lacks usable numbers or loses filter/page context.	Reported; prior URL-cleanup defect requires regression coverage.	Shared pagination.

ID	Issue or concern	Evidence/status	Workstream
WEB-13	Bengali wording is unnatural or mixed with English interface text; time formatting is inconsistent.	Reported.	Translation and date audit.
WEB-14	Authors, categories, series, and reading lists lack useful lists or related-book links.	Reported; include public and personal visibility rules.	Directories and entity details.
WEB-15	Printed-book creation remains exposed despite its removal request.	Confirmed entry points; attempted removal needs verification.	Creation navigation and forms.
WEB-16	Review-required/draft terminology misrepresents the automatic publication workflow.	Reported.	Lifecycle labels and operations.
READ-01	Text is confined to a narrow central column instead of using available screen space.	Confirmed former 800px default; full-width change needs acceptance.	Reader canvas and width preferences.
READ-02	Font-size actions are hidden inside Settings.	Confirmed; toolbar changes need acceptance.	Primary reading controls.
READ-03	Excessive header, toolbar, and pagination content takes space from the book.	Reported and evident in the layout structure.	Reader chrome and fullscreen.
READ-04	Fullscreen shows unwanted persistent controls or lacks intuitive access to tools.	Reported; test idle and deliberately revealed states.	Immersive interactions.
READ-05	Settings and actions lack a complete, discoverable shortcut system.	Confirmed limited command coverage.	Shared command registry.
READ-06	Rich EPUB behavior may differ from the previous reader.	Requires feature-by-feature comparison.	Views, gestures, media, contents, and annotations.
READ-07	Reflow, chapter changes, and replacement editions can threaten exact saved positions.	Required regression area; do not assume every case is defective.	Anchors, CFIs, and progress.
ENG-01	A valid file can still be poorly chaptered or semantically misstructured.	Confirmed gap between text integrity and structural understanding.	Canonical document planning.

ID	Issue or concern	Evidence/status	Workstream
ENG-02	Canonical export deliberately skips late chapter inference.	Confirmed code path; structural inference must happen earlier.	Pre-canonical structuring.
ENG-03	Export verification checks text occurrence without fully proving EPUB spine order.	Confirmed verifier behavior.	Independent order-aware verification.
ENG-04	Contributor extraction relies on heuristics, fixed confidence values, and a few specific corrections.	Confirmed implementation.	Evidence-based entity extraction and resolution.
ENG-05	Source headings, numbering, typography, and contextual boundaries may remain inside one body section.	Corpus investigation candidates; not all single-flow works require splitting.	Chapter intelligence.
ENG-06	Collection failures, unavailable source content, and application defects are hard to distinguish.	Reported; retain real causes and recovery paths.	Classification and recovery.
ENG-07	Administrators cannot easily identify which unfinished works actually changed.	Reported; compare body, structure, and metadata changes.	Update detection and presentation.
ENG-08	“Prepared” can be shown without proving usable HTML, EPUB, and reader resources.	Reported; enforce end-to-end readiness.	Publication gate.
QA-01	Earlier reports and passing layout tests overstate visual or functional acceptance.	Confirmed mismatch between recorded claims and unresolved feedback.	Evidence and completeness reporting.
QA-02	Unreported defects may exist in pages, dialogs, forms, and failure states.	Explicit investigation requirement.	Full inventory audit.

Audit every portion of every page

For each route and dynamic variant, inspect its shell, heading, local navigation, search, filters, result summary, cards/table, actions, forms, dialogs, feedback, pagination, exports, and footer. Mark absent portions as not applicable rather than silently omitting them.

For each applicable portion, inspect populated, loading, empty, no-match, failed, unavailable, stale, saving, success, permission-denied, and disabled states. Include keyboard focus, pointer interaction, touch behavior, translated labels, enlarged text, delayed data, and image failures.

Each issue record must contain reproduction steps, affected route/state, screenshot, expected behavior, root cause, implementation task, regression check, and accepted evidence. Status transitions are **investigate → confirmed → implemented/unverified → verified**. Preserve initial failure evidence.

Previous-version completeness report

Use the pinned previous-version baseline recorded in feature parity, the retained frontend, and the previous embedded reader as references. Record each capability as retained, missing, incomplete, intentionally changed by the user, or verified. Include its current destination and real interaction evidence. Do not use one overall completion percentage to hide feature gaps.

4. Plan A: website and admin experience

A1. Establish one design system

- Use a shared 4/8/12/16/24/32px spacing scale with clear separation between sections and smaller gaps inside related controls.
- Standardize typography, border treatment, surface colors, content widths, control padding, focus rings, and responsive breakpoints.
- Give controls a 44px minimum target. Related toolbar controls must align and have matching normal-size heights. At enlarged text sizes, permit growth without clipping.
- Define primary, secondary, destructive, and icon-only actions. Use consistent icons and accessible names; expose text where an icon alone would be unclear.
- Reuse components for headings, toolbars, filters, cards, tables, tabs, disclosures, forms, dialogs, feedback, exports, and pagination. Remove conflicting overrides rather than adding another patch layer.
- Keep primary content near the beginning of the page. Instructions, exports, technical identifiers, and secondary controls must not dominate it.
- Maintain a calm library presentation with the combined বাংলা পাঠাগার wordmark and consistent active navigation.

A2. Page and section work

Page family	Required audit and resulting behavior
General-user shell	Stable logo/header; My Books, Browse, Notes, and Kindle navigation; account/language controls; mobile safe areas; active section; footer without overlap.
Admin shell	Permission-filtered sidebar, labelled mobile drawer, visible account actions, correct default destination, same-account Reader View, and clear return to Admin.
Home and My Books	Useful signed-in landing, Continue Reading, owned/saved/recent/granted views, clear membership actions, accurate counts, and meaningful empty/error states.
Catalog and search	Compact title/search/sort row, consistent control sizes, optional useful facets, preserved conditions, working clear/reset behavior, and prompt result placement.

Page family	Required audit and resulting behavior
Book cards	Equal heights within a row, consistent cover treatment, readable title/credits, aligned footer actions, saved state, Kindle, and progress inside the card. Do not truncate content merely to force equal height.
Book tables	Readable columns, stable action positions, consistent row density, sensible horizontal scrolling or mobile transformation, and no moving links after account data loads.
Public book details	Prominent cover, title and credits; Read/Continue, Save, Kindle, EPUB, and new-tab HTML; linked metadata; clear progress, notes, and availability.
Personal book details	The same reading hierarchy with account-specific notes/progress and preserved personal-library context.
Other editions	Understandable edition identity, working links, correct access, and preservation of the chosen edition's history.
Authors and contributors	Author-first directory, explicit other roles, clean names, accurate linked-book counts, and working related-book links.
Categories, series, reading lists	Consistent directory/detail layouts, clear terminology, compact search and sorting, meaningful counts, and linked results. Use পাঠতালিকা / Reading Lists for curated lists.
Notes	Four reachable views; aligned search/color controls; consistent book groups and annotation cards; separate editing and exact reader navigation; distinct empty, no-match, and retry states.
Kindle	Discoverable send controls, understandable recipient selection and confirmation, operation details, history, explicit resend, compact filters, and honest uncertain-delivery status.
Profile and security	Coherent identity/photo/password/recipient/security sections, clear validation and feedback, accessible file selection, and conditional two-factor steps.
Login and password flows	Usable labels, validation, pending states, password setup/reset recovery, and two-factor entry only when required by the authenticated flow.
Help, About, Contact	Consistent text hierarchy, usable content width, accurate guidance, working links, and bilingual interface copy.
Physical-book inventory	Preserve existing records, permissions, exports, and personal reading logs. Remove printed-book creation links and creation choices.
Admin overview	Prioritized operational information, real totals, actionable problems, and clear next destinations; avoid an undifferentiated wall of statistics.

Page family	Required audit and resulting behavior
Book Creation/source catalog	Visible Fetch All, Process All, Upload EPUB, and Scheduling; understandable selection scope, real progress, current states, and useful next steps.
Source detail	Clear source identity, linked edition, source structure/update evidence, generated files, reader launch, and relevant recovery controls.
Unfinished Books	Consistent source workspace; clear distinction between source-incomplete works and broken processing; all/updated scopes, last check, actual changes, and regeneration progress.
Managed books	Compact search/table, covers, linked credits/categories, meaningful lifecycle states, direct Edit/Files actions, and no false manual-review requirement.
Manual EPUB import	Coherent file/metadata flow, validation, preparation progress, recoverable failures, and a useful completed-book destination. Never describe a record without ready files as a prepared book.
Book editor, relationships, history	Logical metadata/files/history sections, visible save actions, preserved unsaved forms, linked entities, accurate files, and safe regeneration/edition history.
Creation result	Explain the result and next action before technical identifiers; provide actual file, reading, and management destinations.
Managed entities	Consistent hub/list/create/edit layouts for every entity kind, contributor roles, linked counts, duplicate handling, and provenance.
Processing list and detail	Understandable stages and queue positions, real states, actionable causes, pause/resume/cancel/recovery, and optional technical details. Distinguish queueing completion from book completion.
Schedules	Direct Run Now/settings/history, clear task/scope/timing/enabled state, truthful next run, complete reset behavior, and readable execution outcomes.
Users, access, invitations, mail	Searchable account directory, coherent detail sections, understandable grants and expiry, correct self-edit restrictions, security, invitations, and mail uncertainty.
Legacy and missing pages	Correct redirects with context and permissions preserved; useful not-found/unavailable pages with a next action.

A3. Rules that apply everywhere

- Remove book-type and added/request-date browsing filters, including legacy URLs and saved searches. Scheduling and permission-expiry dates remain functional.
- Use real numbered pagination with boundaries, neighbors, ellipses, totals, and retained valid conditions.

- Place exports after results behind one compact Download list disclosure. Export all matching records, not just the visible page.
- Use natural Bengali reviewed against page purpose. Translate interface labels without translating the source book's text or inventing translated contributor names.
- Keep source extraction checks automatic. Show legitimate processing/waiting states without reintroducing a routine manual-review gate or falsely labelling unfinished work ready.
- Test long titles, many credits, missing metadata/covers, saved-state changes, delayed responses, double submission, stale revisions, and expired permissions.

A4. Website acceptance gate

Every page in the inventory must pass its applicable section/state checks in both languages. At 1366×900, normal catalog results must begin within 320px and a normal admin list must show at least eight ordinary rows where applicable. At 390×844, ordinary browse results must begin within 390px. Exceptional empty/error forms have separate recorded expectations.

No horizontal document overflow, clipped labels, overlapping fixed navigation, inconsistent adjacent control sizes, unequal row cards, inaccessible actions, dead links, or unhandled submission states may remain in the audited inventory. Visual inspection and complete task execution are required in addition to geometry assertions.

5. Plan B: standalone ebook reader

B1. Reading-first layout

- Extract the reader application while retaining working EPUB, security, resource, annotation, and position logic.
- Fill the available viewport with the reading canvas. Default new preferences to full-width content; preserve existing saved choices and keep narrower widths optional.
- Give A–, A+, contents, fullscreen, settings, and search predictable primary controls. Use a labelled tools menu for secondary actions.
- Keep the title, return action, language, and tools compact. Do not use large site navigation, explanatory paragraphs, or permanent pagination blocks as the reading layout.
- In fullscreen, hide persistent chrome. Reveal temporary controls through deliberate pointer/touch/keyboard interaction; do not leave Show Controls, Exit Fullscreen, or page buttons over the book at rest.
- Retain scrolling, single-page, and two-page views. Show progress and chapter information unobtrusively.
- Keep all tools accessible on small screens without an oversized permanent toolbar or controls disappearing beyond the viewport.

B2. Reader feature inventory

Area	Required behavior and edge cases
Opening	Correct prepared edition, cover/title/body order, loading, invalid EPUB, unavailable files, expired access, and retry guidance.
Contents	Nested hierarchy, current chapter, ancestor expansion, collapse controls, keyboard operation, exact navigation, and long titles.
Typography	Font size, family, line spacing, width, theme, and view changes with immediate feedback and persisted preferences.
Page navigation	Real page turns, chapter boundaries, keyboard/touch gestures, orientation changes, correct labels, and no false page totals.
Search	All-chapter search, clear results, exact match navigation, cancellation, no match, and partial/unavailable chapter errors.
Annotations	Accurate bookmarks, highlights, quotations and notes; editing/deletion; selection preservation; exact links from Notes.
Progress	Resume the real passage, retry failed saves, preserve the latest position, and handle multiple tabs/sessions predictably.
Fullscreen	Native fullscreen and focus fallback, discoverable gestures/keys, hidden idle chrome, usable temporary tools, and preserved reading position.
Rich content	Images, poetry, tables, footnotes, internal/external links, audio/video, safe media seeking, and no autoplay.
Local EPUB	Device-only import, offline shell, matching reading controls, malformed archives, resource cleanup, and no accidental upload.
Legacy locations	Restore supported text/spatial/temporal CFIs and edition-bound annotations. Report stale targets without overwriting them.
Accessibility	Consistent names, focus restoration, keyboard reachability, readable themes, enlarged text, reduced motion, and meaningful errors.

B3. One command registry

Toolbar actions, shortcuts, help, translated labels, enabled states, and command handlers must use one registry shared by the library and local readers. Unavailable local-reader/account actions must not appear functional.

Action	Default key
Fullscreen / focus mode	`F`

Action	Default key
Close active panel or leave fullscreen	`Esc`
Contents	`T`
Settings	`S`
Search book	`/`
Shortcut help	`?`
Larger / smaller text	Existing `Ctrl/⌘ +` and `Ctrl/⌘ -`
Paper / sepia / dark theme	Existing `Ctrl/⌘ 1/2/3`
Cycle reading view	`V`; reverse with `Shift+V`
Cycle font	`G`; reverse with `Shift+G`
Cycle width	`W`; reverse with `Shift+W`
Increase / decrease line spacing	`L` / `Shift+L`
Previous / next page	Arrow keys, Page Up/Down, and existing Space behavior
Previous / next chapter	`[` / `]`
Bookmark current position	`B`
Open annotations	`A`
Highlight selected text	`H`

Activate reader commands only in the reader context. Protect form editing, media controls, text selection, and unrelated browser shortcuts. Provide a setting to disable single-character bindings and show current shortcuts beside their actions and in help. Test actual keyboard events in the reading document, toolbar, dialogs, and fullscreen.

B4. Reader acceptance gate

- At normal portrait phone and desktop sizes, at least 80% of viewport height remains available for reading with normal controls shown. Record separate landscape and enlarged-text expectations.
- After fullscreen controls hide, the reading canvas uses at least 98% of the viewport.
- A−/A+ work without opening Settings. Every setting has a discoverable keyboard path and documented shortcut.
- Changing size, font, line spacing, width, theme, view, orientation, or viewport preserves the current passage.

- Contents and search reach the correct chapter/passage. Reader links from Notes restore the verified target.
- Fullscreen tools remain usable without persistent idle buttons or obscured content.
- No authored scripts, unsafe archive paths, unrelated authenticated network requests, or cross-account resources become accessible.
- Local reading remains offline and device-only. Existing library permission and read-once rules remain intact.

6. Plan C: intelligent curation and book creation

C1. Planning-time findings

The read-only census on 21 October 2026 inspected 4,171 source catalog records, including 60 source-marked unfinished works. It counted 27,195 cached source pages and 2,238 curated documents. Processing continues, so these are a moving snapshot rather than a completeness percentage.

Among curated documents, 327 had one body section; 190 such sections exceeded 20,000 HTML characters, and 101 contained heading or bold markup. These need semantic investigation. A single body section is not inherently wrong: some works are genuinely continuous.

Two implementation gaps are confirmed:

1. Canonical exports intentionally skip late chapter inference to protect verified source text. Improved structure must therefore be established **before canonicalization**, not improvised by the exporter.
2. Existing export checks look for body-text occurrences across HTML/XHTML. This does not independently establish correct EPUB spine order, complete section classification, or good chapter boundaries.

C2. Audit the entire source corpus

Freeze a catalog revision and account for every source record. Inspect all relevant reading pages, contents pages, linked components, collected content, and bibliographic evidence. Reuse immutable cached source pages when valid; fetch missing or changed material through the controlled source transport.

For every book, record collection coverage, layout family, hierarchy, source/body flow, front/back matter, credits, ambiguities, incompleteness, update evidence, current output, and discrepancies. Finish with a reasoned classification: understood, structurally ambiguous, source-unavailable, or defective. No book may silently disappear from the audit.

Build a source-grounded 300-book reference corpus spanning every observed layout and difficult case, including novels, anthologies, poetry, drama, linked compilations, long unchaptered pages, nested/paginated contents, mixed credits, and incomplete sources. Use 200 for development and reserve 100 for held-out evaluation. Keep related works/editions in the same split. Engineering review establishes the expected entities, boundaries, and reading order; routine administration does not require per-book review.

C3. Processing stages

Stage	Required result
Discover	Versioned complete catalog, source identity, update scope, and explicit collection status.
Collect	All relevant allowed pages in source order, immutable snapshots, page hashes, and component relationships.
Parse source blocks	Stable IDs and locations for paragraphs, headings, lists, poetry, images, tables, footnotes, and other content.
Understand structure	Source-grounded chapter/section hierarchy, front/body/back roles, component boundaries, and reading order.
Extract entities	Bibliographic entities and contributor roles supported by exact source spans and evidence.
Resolve identities	Canonical identities and aliases with retained evidence; no name-similarity-only merging.
Canonicalize	One versioned document containing blocks, hierarchy, entities, source mappings, and confidence/validation evidence.
Generate	HTML, valid EPUB metadata/navigation/spine, covers, and reader resources from that same document.
Verify	Independent source coverage, entity evidence, chapter/section structure, file contents, reading order, links, and reader readiness.
Publish	Atomic activation after successful checks, with idempotent website handoff and prior edition retained.
Monitor	Material source-change detection, understandable update summaries, safe regeneration, and version-aware recovery.

C4. Intelligence without paid APIs

Use deterministic evidence first: explicit source navigation, labels, typography, numbering, neighboring blocks, and repeated structural patterns. Use local model inference for ambiguous decisions.

The initial benchmark candidate is Qwen3-4B GGUF, Q4_K_M, served privately through llama.cpp. Pin model/runtime revisions and checksums before installation. These are candidates to evaluate, not a claim that this model already meets the required Bengali extraction accuracy.

Default execution policy:

- One inference worker, bounded context windows, overlapping neighboring source context, and a carried-forward outline.

- Structured, schema-validated decisions containing source block IDs and evidence references.
- Cache decisions by source content, extractor, model, prompt, and schema versions.
- No model execution on ordinary website or reader request paths.
- No paid endpoint, free-tier dependency, or automatic cloud fallback.
- Resource preflight and explicit waiting states when memory/disk are insufficient. Available planning-time hardware was 8 GB RAM with approximately 12 GB free disk.
- Benchmark accuracy, peak memory, latency, and projected full-catalog runtime before enabling corpus-wide inference.

The model proposes labels, relationships, and boundaries. It must not rewrite, summarize, translate, invent, or silently discard the book's text. A source-content prompt cannot become an instruction to execute tools, access credentials, or change policy.

C5. Chapter and reading-flow intelligence

- Recognize Bengali/English chapter conventions, numbered boundaries, bold paragraph headings, acts/scenes, poem titles, anthology entries, and nested sections inside long pages.
- Distinguish real chapters from scene breaks, dialogue, dates, credits, ordinary bold prose, repeated navigation, and unrelated recommendations.
- Preserve source hierarchy and component-book order. Linked compilations require independently complete component evidence.
- Preserve body reading order when separating metadata, dedication, prefaces, and back matter.
- Keep genuinely continuous works continuous. Do not divide text merely to increase chapter counts.
- Use original headings where available. Any necessary generated navigation label must be neutral, localized, and separate from source-authored body text.
- Keep every block accounted for. Remove repeated website chrome only with evidence; preserve meaningful repetition inside the work.
- Use stable source-block mappings so segmentation changes do not needlessly invalidate notes and reading positions.

C6. Entity extraction and curation intelligence

- Extract work/edition identity, authors, translators, editors, publishers, illustrators, cover artists, series and sequence, categories, language, original title, edition, and publication facts when supported.
- Handle combined roles such as translation and editing by the same person.
- Distinguish a person credited for the book from someone mentioned in a dedication, introduction, quotation, or narrative.
- Distinguish publisher role from entity type; a publisher need not always be an organization.
- Separate bibliographic facts from prose even when they appear inside the first reading page or an unlabelled block.
- Preserve raw values, exact source spans, role evidence, normalized forms, aliases, and resolution history.
- Do not fabricate missing names or merge people solely through fuzzy name matching.

- Replace isolated special-case corrections with reusable evidence-based rules while retaining already verified corrections and their audit history.
- Calibrate confidence against the held-out reference set. Model self-confidence or hardcoded confidence values alone do not justify publication.

C7. Independent validation and publication

Verification must rebuild expectations from source evidence rather than merely accepting the extractor's own success flag.

- Account for all retained source blocks and detect missing, duplicated, reordered, or incorrectly relocated content.
- Verify EPUB body order by traversing the actual package spine. Navigation documents cannot count as body-text evidence.
- Check section boundaries, front/body/back classification, contents targets, assets, metadata, entity provenance, and exported text.
- Preserve poetry, tables, footnotes, images, internal links, and meaningful repetition.
- Produce valid EPUB navigation without duplicate visible contents pages. Reconcile older documentation against valid format requirements rather than generating invalid packages.
- Require real HTML/EPUB artifacts and prepared reader resources before reporting readiness.
- Activate a replacement atomically only after all required checks pass. Retain the previous valid edition and personal data.
- Publish verified source editions automatically. Structural ambiguity receives automated analysis/recovery, not a blanket admin-review requirement or a fabricated success state.

C8. Updates, faults, and recovery

- Compare source body, hierarchy, credits, covers, and completion status; a table-of-contents comparison alone is insufficient.
- Show last checked time, changed portions, source completion changes, queued rebuilds, and activated editions in the unfinished workspace.
- Keep source unavailability, source-marked incompleteness, uncertain structure, transient transport faults, and application defects distinct.
- Recollect affected pages or rerun affected stages while reusing valid cached work.
- Preserve request IDs, attempt history, checkpoints, stop requests, revision checks, authority checks, and worker fencing.
- Retry availability failures with pacing; do not repeatedly rerun an unchanged application defect or bypass verification to clear it.
- Repair recurring application defects at their cause and add regression fixtures before version-aware recovery.

C9. Engine acceptance gate

Account for the entire frozen catalog and report every unresolved/unavailable record. The reference corpus requires 100% accounted-for retained source content and correct reading order. The held-out set requires at

least 99.5% entity precision and 98% chapter-boundary F1, with role, layout, and language breakdowns and recall reported separately.

Run real local inference for semantic evaluation; mocked responses establish integration behavior only. Record throughput and resource requirements. Do not lower thresholds, hide ambiguous cases, or claim full-catalog correctness from benchmark results alone.

7. Delivery sequence and acceptance gates

Milestone	Deliverables	Exit condition
M0: baseline	Consolidated issue register, previous-version capability matrix, complete route/section/state inventory, current screenshots, and service contracts.	All reported issues mapped; uninspected areas explicitly recorded.
M1: website	Shared components, admin/general shells, every website page, bilingual copy, consistent lists/forms, and completed user/admin workflows.	Website visual and interaction gates pass; no open audited website UX/visual issues.
M2: reader	Independent reader service, reading-first canvas, full controls/shortcuts, rich EPUB behavior, local/offline reading, and preserved positions.	Reader gate and cross-service authorization/navigation checks pass.
M3: corpus understanding	Engine service separation, complete source corpus inventory, structural taxonomy, reference corpus, and local inference benchmark.	Every source record accounted for; benchmark results and runtime limits recorded.
M4: intelligent engine	Evidence-based extraction, canonical structure, file generation, independent verification, updates, and recovery.	Engine quality, integrity, and failure-handling gates pass.
M5: controlled rollout	Shadow comparisons, representative canaries, reversible migration/cutover, affected-book reprocessing, and final completeness report.	Data preservation and actual publication/reading flows verified; no concealed failures.

Verification matrix

- Build, lint, strict type checks, appropriate backend/unit tests, generated schema consistency, migrations, architecture/file-size policies, and whitespace checks.
- Bengali and English at 320, 390, 820, 1366, and 1920px; relevant portrait/landscape layouts and 200% text.
- Chromium, Firefox, WebKit, Android, and iPhone simulation. Keep physical-device and external delivery claims separate from simulator evidence.

- Every applicable page/section state, including long content, delayed hydration, missing images, failed requests, stale revisions, expired access, keyboard focus, and repeated submissions.
- Measured control alignment, spacing, minimum targets, card heights, content budgets, contrast, focus, and layout stability.
- Rendered screenshot inspection and complete task execution. Do not automatically replace reference screenshots to accept a regression.
- Actual reader shortcuts, position preservation, annotations, offline behavior, archive isolation, and protected resource access.
- Real corpus/model evaluation, source coverage, EPUB spine verification, deterministic fixtures, restart/resume behavior, and publication canaries.

Final report

For every capability and issue, report its final behavior, implementation status, accepted evidence, and any remaining limitation. Include source audit totals, unresolved reasons, semantic evaluation results, interface screenshots, service readiness, and data-preservation results. Historical checks and overlapping test runs must not be presented as fresh or unique evidence.

Do not report the application complete while required work, failed checks, or known visual/usability defects remain. Do not claim external Kindle delivery or physical-device behavior from simulations.

8. Complete page inventory

This inventory starts from the 54 current page route templates. Audit all four managed entity kinds—contributors, categories, series, and reading lists—where a dynamic kind appears. Expand each route into the section/state matrix defined above. Also inspect associated dialogs, menus, redirects, not-found/error boundaries, and operational status surfaces.

Area	Page route templates
Entry and catalog	`/` ; `/catalog/`
Public book	`/books/[slug]/` ; `/books/[slug]/editions/`
Contributors	`/contributors/` ; `/contributors/[slug]/`
Categories	`/categories/` ; `/categories/[slug]/`
Series	`/series/` ; `/series/[slug]/`
Reading lists	`/collections/` ; `/collections/[slug]/`
Personal library	`/my-books/` ; `/my-books/[slug]/` ; `/physical-books/`
Personal workspace	`/notes/` ; `/profile/`

Area	Page route templates
Reader	`/reader/` ; `/reader/[identifier]/` ; `/local-reader/`
Kindle	`/kindle/` ; `/kindle/books/[slug]/` ; `/kindle/operations/[identifier]/`
Authentication	`/login/` ; `/create-password/` ; `/reset-password/` ; `/reset-password/confirm/`
Information	`/help/` ; `/about/` ; `/contact/`
Legacy	`/[legacy]/`
Admin entry	`/manage/` ; `/manage/overview/`
Admin books	`/manage/books/` ; `/manage/books/new/` ; `/manage/books/[slug]/` ; `/manage/books/[slug]/relationships/` ; `/manage/book-creations/[identifier]/`
Admin entities	`/manage/entities/` ; `/manage/entities/[kind]/` ; `/manage/entities/[kind]/new/` ; `/manage/entities/[kind]/[identifier]/`
Sources	`/manage/sources/` ; `/manage/sources/[identifier]/` ; `/manage/unfinished/`
Processing	`/manage/jobs/` ; `/manage/jobs/[identifier]/`
Schedules	`/manage/automations/` ; `/manage/automations/new/` ; `/manage/automations/[identifier]/`
Users and access	`/manage/users-access/` ; `/manage/users-access/[identifier]/` ; `/manage/users-access/invite/` ; `/manage/users-access/mail/[identifier]/`

Related evidence

- Previous-version completeness register
- Existing interface implementation report
- Existing sitewide UI review
- Source processing architecture
- Protected reader architecture
- Earlier EPUB pipeline requirements

These references support investigation. Their older completion claims do not close the acceptance gates in this document.